

NORCET Ki POL



Maternity Nursing LMR

High Yield Important Topics

BASIC OBSTETRICS

Obstetric score (GTPAL)
Calculation of EDD (Naegele's rule)
Leopold's maneuvers
Partograph & labour care guide
Pelvis

ANTENATAL CARE

Antenatal care (ANC visits, investigations)
Antepartum screening
Anemia in pregnancy
Anemia Mukht Bharat Program
Diabetes in pregnancy
Pregnancy-induced hypertension (PIH)
Dose and indication of Antenatal periods
Teratogens

OBSTETRIC HEMORRHAGE (VERY HIGH YIELD)

Post-partum hemorrhage (PPH)

- Causes
- Management
- Risk factors

Antepartum hemorrhage

- Placenta previa
- Abruptio placentae

Placenta accreta spectrum

- Accreta
- Increta
- Percreta

PRENATAL DIAGNOSIS

Chorionic villi sampling (CVS)

Amniocentesis

ABORTION & EARLY PREGNANCY COMPLICATIONS

Abortion

MTP Act

Ectopic pregnancy - management

Twins (types, superfecundation, superfetation)

MOLAR PREGNANCY

Hydatidiform mole

Snowstorm appearance

Management & follow-up

INTRAPARTUM COMPLICATIONS

Cord prolapse

Shoulder dystocia - management

Breech presentation & maneuvers

Episiotomy

Active management of 3rd stage of labour (AMTSL)

OPERATIVE OBSTETRICS

Instruments for

Curettage & Caesarean section

GYNAECOLOGY

Contraceptives (Classification, duration, LARC method, Regimen for missed pill, PPIUCD)

Disorders of Menstruation

Screening methods for cancer cervix

Pelvic Inflammatory Disease

Genital fistulae

Uterine Fibroids

Asherman Syndrome

PCOS

Asherman Syndrome

IMAGE BASED QUESTIONS

Different types of Placenta

Fetal heart Rate variations (MOST IMPORTANT)

Obstetrics instruments

Obstetrics maneuver (very high yield)

Contraceptive images

PUERPERIUM

Lochia (Duration and Characteristics)

Duration and Management of Puerperium

Drugs You MUST Memorize (Non-Negotiable)

- Drug
- Indication
- Key Nursing Points

Oxytocin

- Induction,
- PPH
- IV infusion rate, uterine hyperstimulation

MgSO₄

- Eclampsia
- Toxicity signs, antidote

Misoprostol

- PPH
- Route, dose

Methylergometrine

- PPH
- Contraindicated in HTN

Iron sucrose

- Anemia
- Test dose, monitoring

LABOUR (CORE NORCET AREA)

Stages of Labour (WITH EVENTS)

Stage: From → To

Key Nursing Duty

1st

True labour → full dilatation

Partograph

2nd

Full dilatation → birth

Bearing down, fetal heart

3rd

Baby → placenta

AMTSL

4th

First 1-2 hrs

PPH monitoring

HOW TO COVER OBG FOR NORCET

1. Use a Protocol-Based Study Approach

For every condition/situation, train your mind to answer 5 questions:

- Definition / criteria
- Most common cause
- First-line drug
- Nursing priority
- Contraindication / complication

2. Fix a Single Source Rule

- One theory source + one MCQ source only.
- No reference hopping.

Your objective is retention + recall, not reading volume.

Implementation:

Read once → extract exam points → never re-read full text. ONLY YOUR NOTES

3. Convert Theory into Decision Trees

Rewrite information as:

“If X → do Y”

“If complication → first action = ____”

This trains you for:

- Priority-based questions
- Case scenarios

Example method:

- Draw flowcharts instead of paragraphs.

4. Enforce a Same-Day MCQ Rule

MCQs must be solved on the same day you study.

5. Build a One-Page Emergency Sheet

Maintain ONE A4 sheet for:

- Emergency drugs
- Sequences
- Red flags

Update it daily.

- Revise it every 48 hours.

This sheet becomes your last 7-day revision material.

6. Use Reverse Learning

Start from MCQs → go back to theory.

Identify:

- Why the correct option is correct
- Why others are wrong
- Add only exam traps to your notes.

This method aligns with NORCET trap-based options.

7. Train for Nursing Language

While solving MCQs, actively look for:

- "First nursing action"
- "Priority intervention"
- "Most appropriate nursing care"

8. How Questions Are Framed In Norcet

- "First nursing action"
- "Drug of choice"
- "Most common cause"
- "Contraindication"
- "Priority intervention"

9. Do NOT Do These (Exam-Killing Errors)

- Studying gynecology equal to obstetrics
- Ignoring drug doses & contraindications
- Reading without MCQs
- Using multiple textbooks



BASIC OBSTETRICS

Fertilization & Early Development

- Site of fertilization: Ampulla of the fallopian tube.

Timeline After Fertilization

- Day 1-3 :Zygote undergoes cleavage
- 8-16 cell stage :Morula
- Day 4 : Morula enters the uterine cavity
- Day 5 : Zona pellucida disappears • Blastocyst formation begins
- Day 6-10 : Implantation window (best time for implantation)
- Day 11 : Implantation completed (counted as day 25 of menstrual cycle)

Total Duration of Pregnancy

- 40 weeks = 280 days = 9 months + 7 days
- Calculated from the first day of LMP (not from conception)
- Also called:

Period of gestation (POG)

Period of amenorrhea (POA)

ACOG Recommendation

- Pregnancy should be terminated at ≥ 41 weeks
- Induction of labor is recommended beyond this period

Classification of Pregnancy by Gestational Age

Preterm:< 37 weeks

Early term: 37 weeks to 38 weeks + 6 days

Full term: 39 weeks to 40 weeks + 6 days

Late term: 41 weeks to 41 weeks + 6 days

Post term: ≥ 42 weeks

✂ NORCET trap: 37 weeks is term but not full-term.

OBSTETRICAL SCORE

Method 1: Gravida & Para

Gravida (G): Total number of times a woman has been pregnant

- Includes past + present pregnancy
- Twins count as one pregnancy

Para (P): Number of pregnancies reaching viable age

- In India: ≥ 28 weeks

Method 2: Two-Number Parity System

Written as G - P/A

- First number: Parity (births beyond viability)
- Second number: Abortions
(< 20 weeks — includes ectopic & molar)

Definitions:

- Nullipara: No previous birth
- Multipara: ≥ 2 previous births
- Grand multipara: ≥ 4 previous births

Method 3: GTPAL System (NORCET Favorite)

G - Gravida: Total pregnancies (twins = 1)

T - Term births: ≥ 37 weeks

P - Preterm births: 20 to 36 weeks + 6 days

A - Abortions: < 20 weeks (includes ectopic, molar)

L - Living children: Current living offspring

(Twins counted as 2)

✂ NORCET trick:

Twins $\rightarrow G = 1$, but $L = 2$

CALCULATION OF EDD:

For Women With Regular Menstrual Cycles

- Assume a 28-day menstrual cycle unless stated otherwise

Naegle's Formula (Presumptive EDD)

EDD = First day of LMP + 9 months + 7 days

✧ Practical rule:

- Add 7 days first
- Then add 9 months
- Exception: If LMP is in February, calculate carefully due to fewer days

NORCET Tip: EDD is always calculated from LMP, not from conception.

If Menstrual Cycle Is NOT 28 Days

- Adjusted EDD = Presumptive EDD + (Cycle length – 28 days)

Examples:

30-day cycle → add 2 days

26-day cycle → subtract 2 days

When LMP Is Unreliable or Unknown

Situations:

- Irregular menstrual cycles
- Lactating mother
- On oral contraceptive pills
- Unsure or unknown LMP

Best Method to Date Pregnancy

✓ Ultrasonography (USG)

Most Accurate Parameter

✓ Crown-Rump Length (CRL)

✧ NORCET Golden Rule:

The earlier the ultrasound is done, the more accurate the gestational age estimation.

NORCET One-Line Memory Aids

- LMP known → Use Naegele's formula
- LMP unreliable → Use CRL on early USG

LEOPOLD'S MANEUVER (FAQ)

Purpose: To assess fetal lie, presentation, position, attitude, and engagement using systematic abdominal palpation.

NORCET rule to remember

- First 3 maneuvers → face the mother's face
- Last maneuver → face the mother's feet

First Leopold Maneuver - Fundal Grip

How: Both hands palpate the uterine fundus

Assesses: Fetal lie

Second Leopold Maneuver - Umbilical / Lateral Grip

How: Hands placed on both sides of the abdomen

Assesses:

- ✓ Fetal position (left or right)
- ✓ Location of fetal back

Findings:

Fetal back: Smooth, firm, curved, resistant surface

Limbs: Small, irregular, multiple knob-like parts

Interpretation:

Back on left → LOA / LOP / LOT

Back on right → ROA / ROP / ROT

✧ NORCET Pearl: Fetal heart sounds are best heard over the fetal back.

Third Leopold Maneuver - Pawlik's Grip

How: One hand grasps the lower uterine segment just above the symphysis pubis

Assesses:

- ✓ Presenting part
- ✓ Engagement (ballottement)

Findings:

Firm, round, globular → Head → Cephalic presentation

Ballottable mass → Head not engaged

Fixed mass → Head engaged

✧ NORCET Pearl: Engagement = presenting part has entered the pelvic inlet.

Fourth Leopold Maneuver – Deep Pelvic Grip

How: Examiner faces the patient's feet

- Fingers press deeply into lower abdomen

Assesses:

- ✓ Fetal attitude (flexion vs extension)
- ✓ Confirms engagement findings

Findings:

Prominent cephalic prominence on same side as limbs → Flexed head (normal)

Prominence on same side as back → Extended head

✧ NORCET Pearl: This maneuver confirms head flexion or deflexion.



A First maneuver



B Second maneuver



C Third maneuver



D Fourth maneuver

NURSES DUO

BY NURSES, FOR NURSES 

Maneuver	Main Purpose
1st (Fundal)	Lie & presentation
2nd (Lateral)	Position & fetal back
3rd (Pawlik)	Presenting part & engagement
4th (Pelvic)	Attitude & confirmation

PARTOGRAPH AND LABOR CARE GUIDE:

PARTOGRAPH — CORE CONCEPT

Graphical tool to monitor labor progress, maternal status, and fetal well-being.

Purpose :

- Early warning for abnormal labor
- Prevents prolonged labor & complications
- Guides referral & intervention
- Medico-legal record

Stages Of Labor (Cervical Dilatation)

First Stage

- Latent phase: 0-3 cm (not plotted)
- Active phase:

WHO modified: ≥ 4 cm
WHO 2018 (LCG): ≥ 5 cm
ACOG: ≥ 6 cm

WHO MODIFIED PARTOGRAPH (VERY IMPORTANT)

Plotting starts:

- From 4 cm onwards (active labor only)

KEY LINES ON PARTOGRAPH

ALERT LINE

- Starts at 4 cm $\rightarrow$ full dilatation
- Cervical dilatation rate: 1 cm/hour
- Minimum acceptable progress
 - $\rightarrow$ Left of alert line $\rightarrow$ Normal labor
 - $\rightarrow$ Right of alert line $\rightarrow$ Slow progress (assess cause)

ACTION LINE

- Drawn 4 hours to the right of alert line
- Crossing = Immediate intervention required

Possible actions:

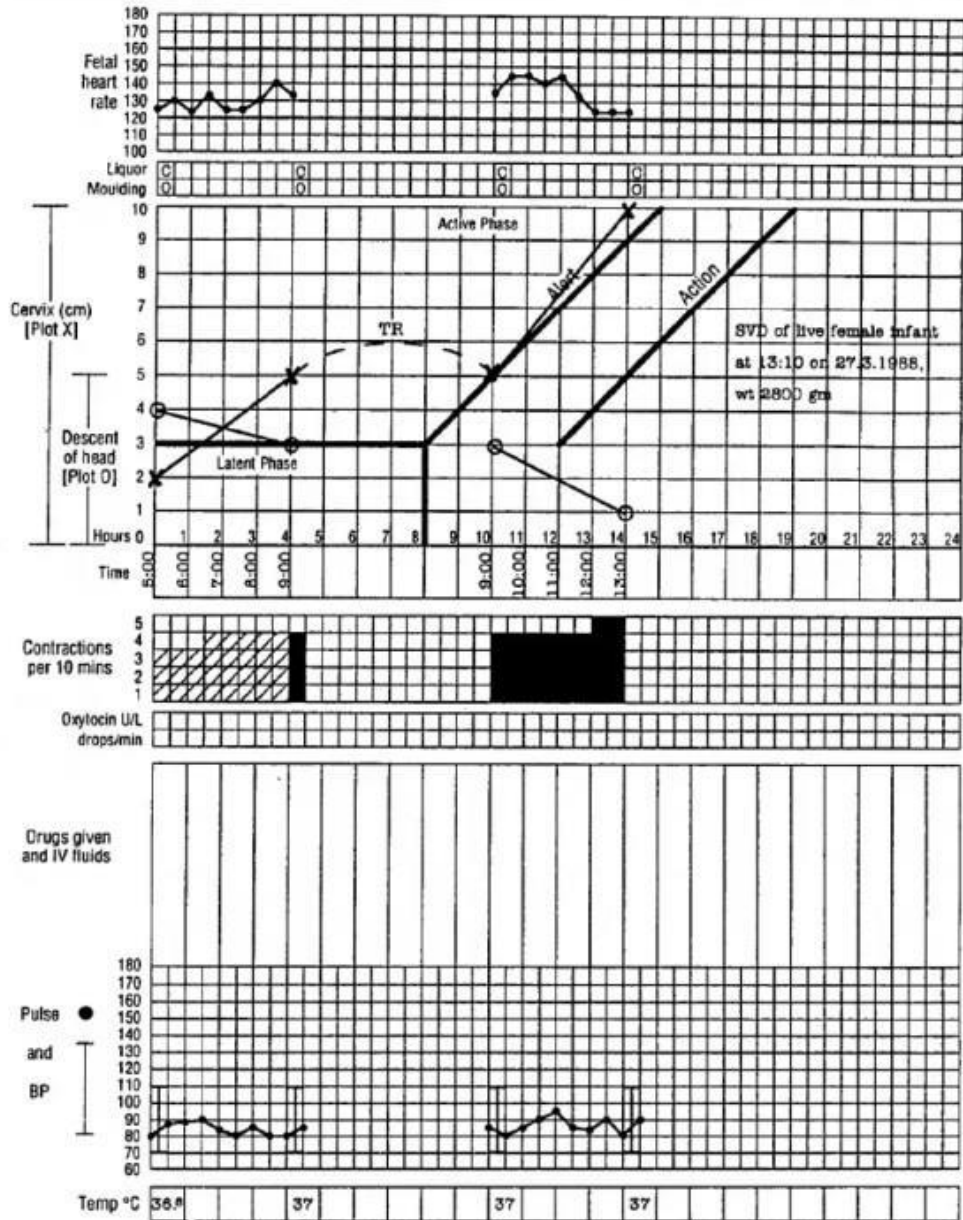
- Augmentation
- Operative delivery
- Referral

GOLDEN EXAM PEARLS ☆

- First cervical dilatation is always plotted on alert line
- Alert line = Transfer line
- Action line crossed = Do not wait
- Used mainly for low-risk pregnancies

PARTOGRAPH

Name Mrs B. Gravida 1 Para 0 Hospital no. 1059
 Date of admission 27.3.1988 Time of admission 8:00 Ruptured membranes 2 hours



FRIEDMAN CERVICOGRAPH

- Classical S-shaped curve
- Not preferred now

**WHO NEXT-GEN PARTOGRAPH/
WHO LABOUR CARE GUIDE (LCG)**

Key shift:

- No Rigid time rules
- Individualized, woman-centered care

WHEN TO START WHO LCG

- Active first stage of labor
- Cervical dilatation ≥ 5 cm
- Low-risk women

STRUCTURE OF WHO LABOUR CARE GUIDE

7 SECTIONS

Section 1: Admission & labor characteristics

Section 2: Supportive care (pain relief, mobility, fluids)

Section 3: Care of the baby (FHR, liquor, moulding)

Section 4: Care of the woman (BP, pulse, temp, urine)

Section 5: Labor progress (dilatation, descent, contractions)

Section 6: Medications (oxytocin, IV fluids, analgesics)

Section 7: Shared decision-making & informed consent

NURSES DUO
BY NURSES, FOR NURSES 

WHO LABOUR CARE GUIDE											
Section 1		Name		Parity		Labour onset		Active Labour diagnosis (Date)			
		Registered midwife (Date)		Time		Risk factors					
Alert column		Time									
		Place									
		ALERT		ACTIVE FIRST STAGE						SECOND STAGE	
Section 2		Supportive care		Contraction		N					
		Pain relief		N							
		Oral fluid		N							
		Position		SP							
Section 3		Maternal		Gestational (Wk)		<30, >35					
		FHR observation		L							
		Amniotic fluid		M+++ , 0							
		Fetal position		B/T							
		Cord		++ +							
Section 4		Woman		Pulse		<90, >120					
		Systolic BP		<80, >160							
		Diastolic BP		<90							
		Temperature °C		<35.0, >37.5							
Section 5		Labour progress		Contraction per 10 min		<2, >5					
		Duration of contractions		<20, >60							
		Cervix (Plat 4)		10							
				9		>2h					
				8		>2.5h					
				7		>3h					
				6		>5h					
		Descent (Plat 0)		5		>6h					
				5							
				4							
3											
2											
Section 6		Medication		Opioid (Plat 0)							
		Medicine									
		Fetal ab									
Section 7		Shared decision making		Assessment							
		Plan									
		Initials									

INSTRUCTIONS: ONLY REPLY OBSERVATIONS PERTAINING TO THE CRITERIA IN THE ALERT COLUMN, ALERT THE NIB OR NISWIFT OR DOCTOR WHO RECORDED THE ASSESSMENT AND ACTION TAKEN IF LABOUR EXTENDS BEYOND TEN HOURS. PLEASE CONTINUE ON A NEW LABOUR CARD WHEN.

Abbreviations: V - Ve; M - Me; B - Benzyl; U - Unknown; SP - Super; MD - Methyl; F - Fatty; C - Cap; W - Wax; E - Ester; A - Alcohol; P - Protein; F₂ - Fatty acid; F₃ - Protein; A₂ - Amino

- Active labor (WHO LCG): ≥ 5 cm
- Alert line slope: 1 cm/hour
- Action line: 4 hours after alert line

- Left of alert line = Normal
- Cross action line = Immediate action

4 cm = alert/action lines
= OLD WHO



WHO MODIFIED PARTOGRAPH (OLD)

- ✓ Active labor starts at 4 cm
- ✓ **Alert line, Action line** present
- ✓ Plot vs. standard **lines: alert, action**
- ✓ Still used in **EXAMS**

5 cm = no lines = LCG



WHO LABOUR CARE GUIDE (NEXT-GEN)

- ✓ Active labor starts at **≥ 5 cm**
- ✓ **NO Alert line, NO Action line**
- ✓ Individual **assessment** vs. lines
- ✓ **NEW WHO** recommendation



NURSES DUO

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PARTOGRAPH - CORE CONCEPT

Definition: Graphical tool to monitor
 ✓ labor progress; maternal status, and fetal well-being

Why used (Exam Focus):

- ⚠ Early warning for abnormal labor
- Prevents **prolonged labor & complications**
- 📄 Guides referral & intervention
- 📁 Medico-legal record



WHO MODIFIED PARTOGRAPH

Plotting starts: From 4 cm onwards
 (active labor only)



KEY LINES ON PARTOGRAPH

- ✓ **Alert Line** : Starts at 4 cm → full dilatation → 1 cm / hour
- ➡ **Left of alert line** → **Normal labor**
- ➡ **Right of alert line** → **Slow progress** (assess cause)



WHO MODIFIED PARTOGRAPH

Plotting starts: From 4 cm onwards
 (active labor only)



ACTION LINE

- ✓ Drawn 4 hours to the right of alert line
- ✓ Crossing = Immediate intervention required



KEY LINES ON PARTOGRAPH

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GOLDEN EXAM PEARLS

- ★ First cervical dilatation is **always** plotted on **alert line**
- ★ Alert line = **Transfer line**
- ★ Action line crossed = **Do not wait**
- ★ Used mainly for **low-risk pregnancies**



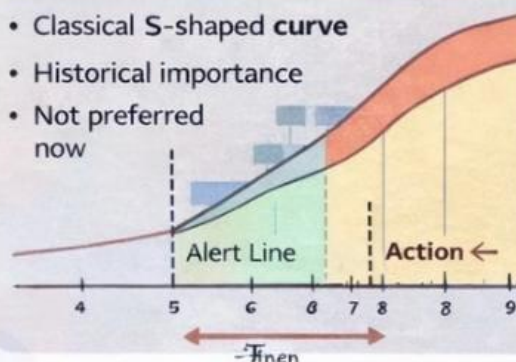
GOLDEN EXAM PEARLS

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FRIEDMAN CERVICOGRAPH

- Classical S-shaped **curve**
- Historical importance
- Not preferred now



WHO NEXT-GEN PARTOGRAPH

WHO LABOUR CARE GUIDE (LCG)

- ✓ Key shift:
- ✓ 🕒 Rigid time rules
- ✓ 🛡 Individualized, woman-centered care
- ✓ 👤 Low-risk women 📄 📱 📺 📺
- ✓ Structure of WHO Labour Care Guide



NCLEX RAPID RECALL

- ✓ Active labor (WHO LCG): ≥ 5 cm
- ✓ **Alert line** slope: 1 cm/hour
- ✓ Action line: 4 hours after alert line
- ✓ Left of alert line = **Normal**

PELVIS:

Key Rule for Identification

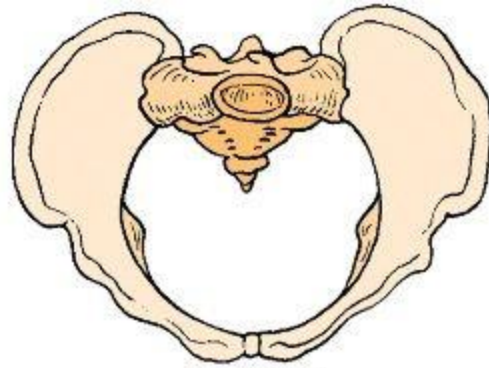
- Pelvic type is identified by the shape of the pelvic inlet (brim)
- Do NOT include coccyx while tracing the inlet

Classification:

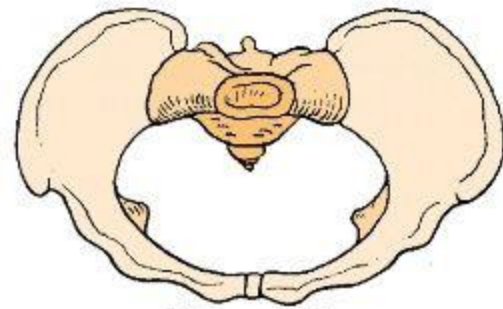
Pelvis Type	Inlet Shape	Key Point
Gynecoid	Round	Most favorable
Android	Heart-shaped	Narrow outlet
Anthropoid	Long oval	OP delivery
Platypelloid	Flat	Poor engagement

NURSES DUO

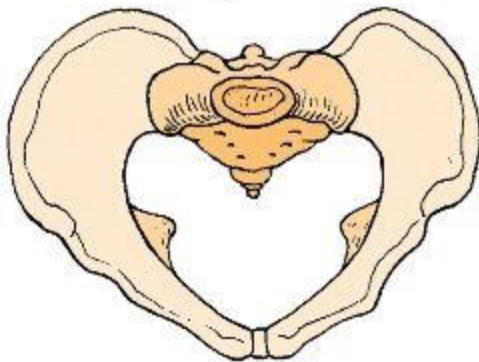
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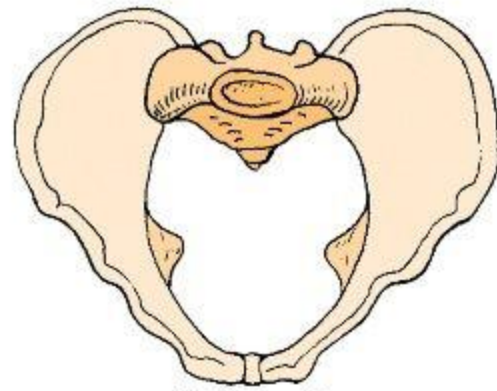
Gynecoid



Platypelloid



Android



Anthropoid

NURSES DUO

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Pelvis Classification

(Caldwell—Moloy)

NCLEX High-Yield Cheatsheet

- ✓ Identify pelvic type by the **shape** the **pelvic inlet (brim)***
- ✓ **No Coccyx** included



No Coccyx

- ✓ G-A-A-P
- ✓ Gynecoid → Good
- ✓ Android → Arrest
- ✓ Anthropoid → AP > T
- ✓ Platypelloid → Problem pelvis

✓ Gynecoid Pelvis



- ✓ **ROUND** shape
- Most common pelvis in females
- ✓ Transverse > AP diameter
- ✓ Best for vaginal birth

NCLEX Pearl: Ideal pelvis for childbirth

NCLEX Pearl:

- Ideal pelvis for childbirth
- Quick **NCLEX** comparison Table

✓ Android Pelvis



- ✗ **HEART-shaped**
- ✓ Transverse > AP diameter
- ✓ Prominent ischial spines
- ✗ Arrest of descent

NCLEX Pearl: Associated w/ difficult labor

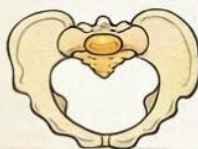
Anthropoid Pelvis



- ✓ **OVAL / Long**
- ✓ AP > Transverse
- ✓ Face-to-pubes delivery common

NCLEX Pearl: OP delivery is common but often successful

✗ Platypelloid Pelvis



- ✗ **PLAT** Oval shape
- Least common
- ✓ Transverse >> AP
- ✗ Poor engagement

NCLEX Pearl: Face presentation common

✗ Platypelloid Pelvis



- ✗ **FLAT** Oval shape
- Least common
- ✓ Transverse >>> AP
- ✗ Face presentation

NCLEX Pearl: Often associated w/ labor dystocia



ANTENATAL CARE

PRECONCEPTION & EARLY PREGNANCY:

Folic Acid Supplementation

All women (low risk): Dose: 400 mcg/day

- Start: 1 month before conception
- Continue: Till 3 months after conception
- Purpose: Prevention of neural tube defects (NTDs)

High-risk (previous baby with NTD):

- Dose: 4 mg/day
- Start: 3 months before conception
- Continue: Till 3 months after conception
- Purpose: Prevention of recurrence of NTD

Special situations

- Folic acid deficiency → 1 mg/day
- Diabetes → 400 mcg/day

On antiepileptic drugs:

- Before conception → 400 mcg/day
- After conception → 4 mg/day

Sickle cell anemia → 5 mg/day

ANTENATAL VISIT SCHEDULE

Up to 28 weeks: 1 visit every 4 weeks

28-36 weeks: 1 visit every 2 weeks

≥36 weeks: Weekly visits

WHO minimum: 8 visits

India (GOI) minimum: 4 visits

Schedule of Antenatal Visits (IMPORTANT)

As per MoHFW guidelines: 4 visits First visit - 12 weeks Second visit - 14-26 weeks Third visit - 28-34 weeks Fourth visit - >36 weeks	As per WHO guidelines: 8 contact visits 1st trimester - 1 contact 2nd trimester - 2 contact 3rd trimester - 5 contact	As per ACOG and what is ideal/ practiced in most settings: 12-14 visits Up- to 28 weeks - every month 28th to 36 weeks - every 2 weeks >36 week -every week
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WEIGHT GAIN IN PREGNANCY

- Normal BMI: 11-12.5 kg
- Low BMI: 12.5-18 kg
- Obese (BMI >30): 7-9 kg

Additional Caloric Requirement

International :

- 1st trimester → 0 kcal
- 2nd trimester → +350 kcal/day
- 3rd trimester → +450 kcal/day

Nutritional Requirements

- Iodine: 250 mcg/day
- Calcium: 1000 mg/day

BMR in pregnancy: ↑ by 10–20%

Fluid retention: ~6.5 L

VACCINATION IN PREGNANCY - HIGH YIELD

Live vaccines contraindicated:

- Measles, Mumps, Rubella
- Varicella (chickenpox)
- Smallpox
- BCG

- HPV

Gap after live vaccine before conception: 1 month

Td vaccine: Given to all pregnant women

- 1st dose at first visit
- 2nd dose after 4 weeks
- If fully immunized in last 3 years → Booster only

MEMORY PEARLS

- Folic acid prevents NTD
- High risk = 4 mg
- 0 extra calories in 1st trimester
- Live vaccines NEVER in pregnancy
- Weekly visits after 36 weeks

ANTEPARTUM (ANTENATAL) SCREENING

First Prenatal Visit (Booking Visit)

- Basic labs (for all pregnant women):
- ABO & Rh typing
- Hemoglobin / CBC → screen anemia
- Urine routine & microscopy (each trimester)
- Urine culture → asymptomatic bacteriuria
- VDRL / RPR → syphilis
- HBsAg → hepatitis B
- HIV testing → opt-out approach
- Rubella IgG → immunity status

High-risk (DM, sickle cell) → Urine culture & sensitivity mandatory

OPTIONAL / INDICATED (If available or risk-based)

- TSH → thyroid screening
- Aneuploidy screening
- First trimester: NT scan + PAPP-A + β -hCG

- Second trimester: Quad test

If Mother Is Rh-negative

- Partner (father) Rh status
- Indirect Coombs Test (ICT)
- At first visit
- Repeat at 28 weeks
- Anti-D prophylaxis if unsensitized

24-28 WEEKS

- Gestational diabetes screening
- 50 g oral glucose challenge
- Repeat Hb

35-37 WEEKS

- Group B Streptococcus (GBS) screening
- Recto-vaginal swab
- Positive → intrapartum IV antibiotics

NORCET PEARLS

- Urine culture is mandatory even if asymptomatic
- Rh-negative care = ICT + Anti-D
- ≥ 3 -week fundal height mismatch → USG first

BY NURSES, FOR NURSES 
ANEMIA IN PREGNANCY

Most Important Concepts

- Most common anemia in pregnancy: **Physiological anemia**
 - Due to plasma volume ↑ more than RBC mass
 - Normocytic, normochromic
- Most common pathological anemia: Iron Deficiency Anemia (IDA)
- Major indirect cause of maternal mortality (India): Anemia

Types Of Microcytic Anemia (EXAM FAVORITE)

Iron Deficiency Anemia (IDA)

Microcytic, hypochromic

NESTROFT: Negative

Management: Oral iron

IV iron (if intolerance / severe)

Blood transfusion if very severe

Thalassemia

- Microcytic, hypochromic
- NESTROFT: Positive
- Management: Blood transfusion (iron NOT useful)

ANEMIA MUKT BHARAT (PREVENTION PROGRAM) High Yield

- Goal: Prevent anemia (not treatment)

Key Interventions:

- Digital hemoglobin estimation
- Iron-Folic Acid (IFA) supplementation
- Deworming
- Iron-fortified foods
- Delayed cord clamping (↓ neonatal anemia)
- Treat other causes (malaria, infections)

Prevention Strategy:

- All reproductive-age females
- 1 IFA tablet weekly
- Albendazole 400 mg twice yearly
- During pregnancy
- Up to 12 weeks: Folic acid only
- From 2nd trimester onward: Iron + Folic acid
- Standard IFA tablet contains:
 - Iron 60 mg
 - Folic acid 500 mcg

WHO SEVERITY CRITERIA

- Severe anemia: Hb < 7 g/dL
- Very severe anemia: Hb < 4 g/dL

MANAGEMENT OF IDA IN PREGNANCY

- Hb < 5 g/dL OR signs of heart failure → Immediate blood transfusion
- Hb ≥ 5 g/dL AND stable → Oral IFA (2 tablets/day)

QUICK PEARLS

- Physiological anemia = dilutional
- IDA = most common treatable cause
- Blood transfusion only when life-threatening

DIABETES IN PREGNANCY

Classification

Pregestational Diabetes (PGDM)

- Diabetes present before pregnancy
- Includes Type 1 & Type 2
- Persists after delivery
- High risk of congenital anomalies

Gestational Diabetes (GDM)

- First detected during pregnancy
- Usually diagnosed at 24-28 weeks
- Resolves after delivery
- NO increased risk of congenital malformations

PREGESTATIONAL DIABETES (PGDM)

Diagnosis

- Any one of the following:
 - Fasting glucose ≥ 126 mg/dL
 - Random glucose ≥ 200 mg/dL
 - 2-hr post-prandial ≥ 200 mg/dL

HbA1c $\geq$ 6.5%

✧ Risk predictor: $\uparrow$ HbA1c

Management of PGDM

- Insulin = drug of choice
- Diet + weight counseling
- Low-dose aspirin $\rightarrow$ prevents PIH

Preconception care:

- Switch OHA $\rightarrow$ insulin
- Folic acid 400 mcg/day
- Delivery: at 39 weeks
- Mode: Vaginal preferred
- C-section if estimated fetal weight $\geq$ 4.5 kg

GESTATIONAL DIABETES (GDM)

Key Features

- No diabetes before pregnancy
- Detected at 24-28 weeks
- No congenital malformations
- Usually resolves postpartum

Diagnosis - DIPSI Guidelines

Non-fasting OGTT

75 g glucose in 300 mL water

Measure 2-hour plasma glucose

- Results
 - < 140 mg/dL $\rightarrow$ Normal (repeat at 24-28 weeks)
 - 140-199 mg/dL $\rightarrow$ GDM
 - ≥ 200 mg/dL $\rightarrow$ Overt diabetes (PGDM)

✧ Test done twice

- First ANC visit

- Again at 24–28 weeks (≥ 4 –week gap)

IADPSG / ADA CRITERIA

Test timing: 24–28 weeks

Method:

- Fasting (8 hours)
- Give 75 g oral glucose
- Measure plasma glucose

Diagnostic Cut-offs

- Fasting : 92mg/dl
- 1-hour : 180mg/dl
- 2-hour : 153mg/dl

☐ If ANY ONE value abnormal $\rightarrow$ GDM diagnosed

MANAGEMENT OF GDM (STEPWISE - LOGIC)

✓ Step 1: First 1-2 Weeks

- Medical Nutrition Therapy (MNT)
- Diet modification
- Weight management
- Physical activity (as tolerated)

🎯 TARGET GLYCEMIC GOALS

(If ALL targets met $\rightarrow$ continue MNT)

- Fasting glucose: < 95 mg/dL
- 1-hour post-meal: < 140 mg/dL
- 2-hour post-meal: < 120 mg/dL
- HbA1c: $< 6\%$
- Average capillary glucose: < 100 mg/dL

📌 Monitoring

- Check post-prandial glucose
- Weekly (diet-controlled)
- Every 2 weeks (on treatment)

✓ Step 2: If Targets NOT Met

- **Start INSULIN (Drug of choice in pregnancy)**
- Oral hypoglycemic agents (metformin, glyburide)
(Only if patient refuses insulin)

Contraindicated in PGDM (NORCET trap)

COMPLICATIONS OF DIABETES IN PREGNANCY

MATERNAL

- Recurrent infections
- Asymptomatic bacteriuria
- Candidiasis
- Polyhydramnios
- Placental enlargement → PIH

↑ Risk of:

- PROM
- Preterm labor
- Cord prolapse
- Abruption placentae
- ↑ Risk of Type 2 diabetes later in life

FETAL

- Fetal hyperglycemia
- Macrosomia (most common)
→ Shoulder dystocia
- Abortion, IUFD, stillbirth
- Diabetic vasculopathy
→ Oligohydramnios
→ IUGR

NEONATAL

- Neonatal hypoglycemia (most common)
- Hypocalcemia
- Hypomagnesemia
- ↓ Surfactant
→ Respiratory Distress Syndrome
→ Necrotizing enterocolitis

- Polycythemia

✧ NORCET Pearl:

Infant of diabetic mother = big baby + low sugar

NORCET QUICK REVIEW

- First line = Diet
- Drug of choice = Insulin
- Most common fetal issue = Macrosomia
- Most common neonatal issue = Hypoglycemia
- Any one abnormal OGTT value = GDM

HYPERTENSION IN PREGNANCY

Definition

- BP $\geq$ 140/90 mmHg on 2 readings $\geq$ 4 hrs apart
- Severe HTN: $\geq$ 160/110 $\rightarrow$ treat immediately
- Occurs after 20 weeks of gestation

Types of Hypertension in Pregnancy

1 Chronic HTN

- ✓ Present before pregnancy or $<$ 20 weeks
- ✓ Persists $>$ 12 weeks postpartum
- ✓ If proteinuria/end-organ damage develops $\rightarrow$ Superimposed pre-eclampsia

2 Gestational HTN

- ✓ HTN after 20 weeks
- ✓ No proteinuria
- ✓ Resolves within 12 weeks after delivery

3 Pre-eclampsia

- ✓ HTN + proteinuria OR end-organ damage
- ✓ Proteinuria Criteria
 - $\geq$ 300 mg/24 hr urine

- Protein : Creatinine ≥ 0.3
- Dipstick $\geq +1$
- Gold standard: 24-hour urine protein

✓ End-Organ Damage (Any ONE)

- Platelets <1 lakh
- $\uparrow$ Liver enzymes ($2\times$ normal)
- Serum creatinine >1.1 mg/dL
- Pulmonary edema
- Cerebral / visual symptoms

Severity of Pre-eclampsia

Without severe features (Mild)

- BP 140-159 / 90-109
- No end-organ damage

With severe features

- BP $\geq 160/110$
- End-organ damage present

⚠ Amount of proteinuria does NOT define severity

Impending Eclampsia - Warning Signs

- Severe headache (not relieved)
- Visual disturbances
- Epigastric/RUQ pain
- Clonus

ECLAMPSIA

Pre-eclampsia + tonic-clonic seizures

Types

- Antepartum (worst prognosis)
- Intrapartum
- Postpartum (≤ 48 hrs after delivery)

Drug of Choice (DOC)

- ✓ Seizure prevention/treatment

Magnesium sulfate

Antidote: Calcium gluconate

- ✓ Acute BP control

IV Labetalol

IV Hydralazine

Oral Nifedipine

- ✗ Contraindicated: ACE inhibitors, ARBs

Magnesium Sulfate - Key NORCET Points

- Not an antihypertensive
- Therapeutic level: 4-7 mEq/L

Toxicity signs:

- Loss of DTR (first sign)
- RR <12/min
- Urine output <30 mL/hr

HELLP Syndrome (HIGH YIELD)

- Hemolysis
- Elevated liver enzymes
- Low platelets (<100,000)

Management:

- MgSO_4 + BP control + delivery
- Definitive Treatment

Delivery of fetus & placenta

Severe PE / eclampsia → Deliver irrespective of gestational age

MEMORY LINE

" MgSO_4 for seizures, Labetalol for BP, Delivery is cure."

TERATOGENS

What are Teratogens?

Any agent causing fetal abnormality when exposure occurs during pregnancy

Incidence of anomalies: ~2-3%

Critical periods:

- Pre-implantation (0–2 weeks) → All or none
- Embryonic (3–8 weeks) → Organogenesis → major malformations
- Fetal period (>8 weeks) → Growth & functional defects

Major Teratogens & Defects:

Anti-Epileptic Drugs

- Valproate → Highest risk
Neural tube defects, orofacial clefts, cardiac defects
- Phenytoin / Carbamazepine → Fetal hydantoin syndrome
Mid-face hypoplasia, distal digital hypoplasia

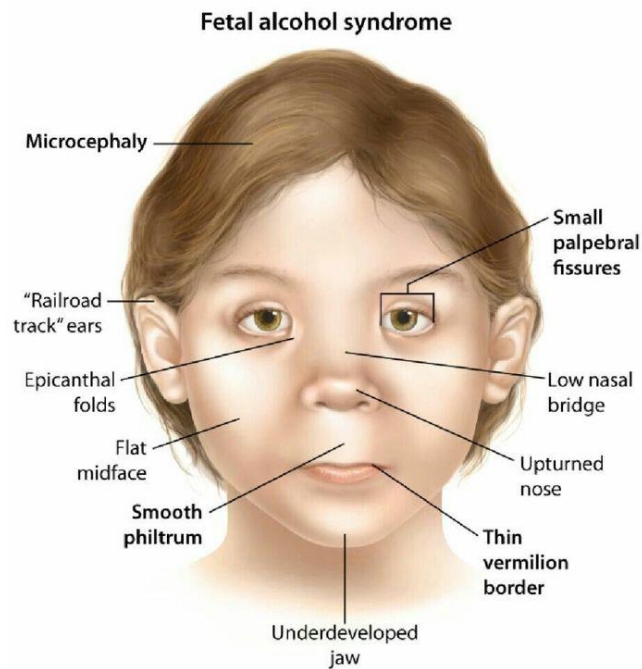
Safer options → Lamotrigine, Levetiracetam



Alcohol

Fetal Alcohol Syndrome (FAS)

- Neurodevelopmental delay
- Facial features: smooth philtrum, thin upper lip, short palpebral fissures
- Most common preventable cause of developmental disability



ACE Inhibitors / ARBs

ACE-inhibitor fetopathy

- Renal hypoperfusion → oligohydramnios
- Pulmonary hypoplasia (Potter sequence)
- Management: Switch to safer antihypertensive before conception

NSAIDs (Indomethacin)

- >32 weeks → Premature closure of ductus arteriosus → pulmonary HTN
- <32 weeks → Relatively safer

Antimicrobial Drugs

- Aminoglycosides → Ototoxicity, nephrotoxicity
- Tetracyclines → Yellow-brown teeth, bone deformities
- Sulfonamides → Neonatal hyperbilirubinemia
- Nitrofurantoin (1st trimester) → Cleft lip, hypoplastic left heart (rare)

Recreational Drugs

- Amphetamines → Major teratogen
- Opioids → Neonatal abstinence syndrome
- Marijuana → Preterm birth, FGR
- Smoking → FGR, APH, preterm labor, abortion

Drug	Key Defect
Methimazole	Cutis aplasia
Warfarin	Stippled epiphyses, nasal hypoplasia (6–9 weeks)
Thalidomide	Phocomelia
Methotrexate	Cloverleaf skull, craniosynostosis
Misoprostol	Moebius sequence
Lithium	Ebstein anomaly
Isotretinoin	Microtia/anotia, flat nasal bridge, neuro defects
DES	Clear cell vaginal carcinoma, T-shaped uterus

Cutis aplasia: METHIMAZOLE



PHACOMELIA: THALIDOMIDE (PYQ)



Di sala Syndrome: WARFARIN



NORCET PEARLS ☆

- ✓ Valproate = highest teratogenic risk
- ✓ ACE inhibitors → oligohydramnios
- ✓ Indomethacin unsafe after 32 weeks
- ✓ Isotretinoin = absolute contraindication
- ✓ Alcohol = preventable cause of disability



OBSTETRIC HEMORRHAGE

POSTPARTUM HEMORRHAGE

Definition

- Vaginal delivery: ≥ 500 mL blood loss
- Cesarean section: ≥ 1000 mL blood loss
- ACOG: ≥ 1000 mL, regardless of delivery mode

Types

- Primary PPH: Within 24 hours
- Secondary PPH: 24 hours-6 weeks

Most Common Causes

- Overall: Uterine atony
- Primary PPH: Atonic uterus
- Secondary PPH: Retained placental tissue

INITIAL MANAGEMENT (FIRST PRIORITY)

[1] Resuscitation (NORCET = ABCs first)

- Two large-bore IV cannulas (14-16G)
- Oxygen via mask
- Monitor urine output

Blood investigations:

- ABO & Rh typing
- Coagulation profile
- Serum fibrinogen
- Platelets & electrolytes

[2] Uterine Assessment

- Per-abdomen exam → Assess uterine tone
- Uterine massage → First-line for atony (FAQ)
- Per-vaginal exam → Check for retained tissue or trauma

ATONIC PPH — STEPWISE MANAGEMENT

[1] Uterotonics (First-line drugs)

- Oxytocin: 10 IU in 500 mL NS (WHO)

- Methylergometrine: 0.2 mg IM (repeat every 2-4 h)
- Carboprost (15-methyl PGF_{2a}): 250 mcg IM every 15-90 min
Max: 8 doses
- Misoprostol (add-on): WHO: 800 mcg oral/sublingual
ACOG: 600-1000 mcg oral or PR

2 **Antifibrinolytic**

- Tranexamic acid
1 g IV in 100 mL NS
Over 10-20 min
Within 3 hours of PPH onset

◆ IF BLEEDING CONTINUES

3 Mechanical Measures

- Bakri balloon tamponade
- Max capacity: 500 mL

4 Compression Sutures

- B-Lynch (most common)

5 Surgical Devascularization

LAST RESORT (LIFE-SAVING)

- ✓ Stable patient → Internal iliac artery ligation
- ✓ Unstable patient → Subtotal hysterectomy

☆ **EXAM PEARLS**

- ✓ Uterine atony = most common cause of PPH

Uterine massage comes before drugs (FAQ)

- ✓ Oxytocin = first-line uterotonic
- ✓ Tranexamic acid must be given within 3 hours
- ✓ Bakri balloon before hysterectomy

ANTEPARTUM HEMORRHAGE

Definition: Vaginal bleeding occurring after periods of viability till delivery.

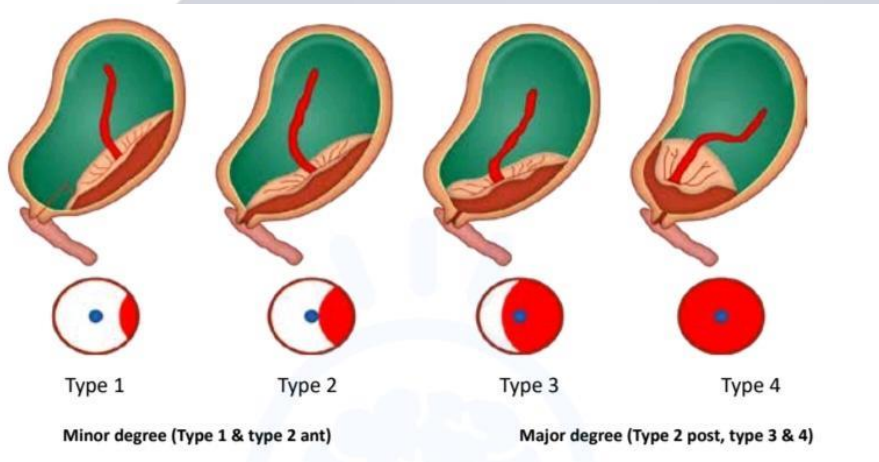
Main Causes:

- Placental (Most Common): Placenta previa, Abruptio placentae, Vasa previa.
- Extraplacental: Cervical or vaginal causes (polyps, trauma).
- Indeterminate

PLACENTA PREVIA

Definition: Placenta implants over or very near the internal cervical os.

Key Risk Factors: Previous C-section or uterine surgery, multiparity, advanced maternal age, smoking, multiple gestation.



Classic Presentation: Sudden, painless, bright red vaginal bleeding, often recurrent.

Management:

✓ **Before Viability/Stable:** Expectant management (bed rest, monitoring, corticosteroids for lung maturity if <34 weeks, RhoGAM if Rh-negative).

✓ **Definitive Treatment:** Cesarean delivery is almost always required.

NORCET Focus:

- Painless bleeding = think placenta previa.
- Do NOT perform digital cervical exams (can cause catastrophic bleeding).
- Delivery is by C-section

ABRUPTIO PLACENTAE

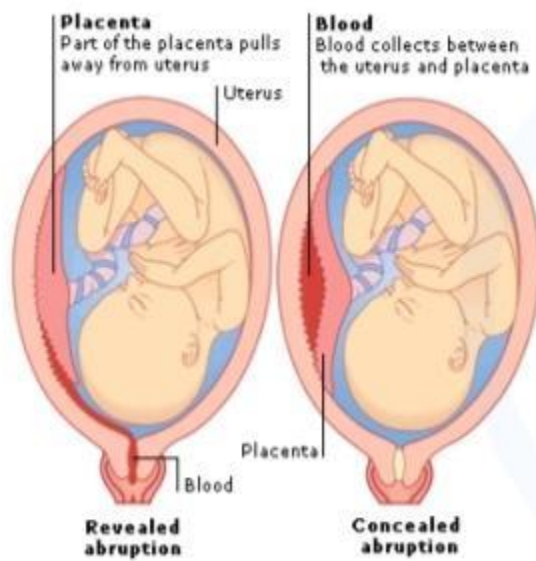
Definition: Premature separation of a normally implanted placenta.

Key Risk Factors: Hypertension (preeclampsia, chronic HTN), trauma, cocaine use, smoking, prior history

Classic Presentation: Painful (abdominal pain), dark vaginal bleeding, uterine tenderness/rigidity, fetal distress. Bleeding may be concealed.

Types:

- Revealed (visible bleeding),
- Concealed (blood trapped behind placenta; more severe, higher risk of DIC).



Complications: Maternal shock, DIC, renal failure, fetal demise.

Management: Immediate delivery (often emergency C-section) is the goal.

- Treat shock and coagulopathy.

NORCET Focus:

Painful bleeding + rigid/tender uterus = think abruption. Monitor for signs of DIC (bleeding from IV sites, petechiae).

VASA PREVIA

Definition: Fetal umbilical vessels traverse the membranes unsupported, crossing the cervical os.

Risk: Associated with velamentous cord insertion or succenturiate placental lobe.

Classic Presentation: Fetal hemorrhage. Rupture of membranes is followed by sudden vaginal bleeding with fetal bradycardia or sinusoidal heart pattern.

Management: Emergency cesarean delivery.

NORCET FOCUS

Bleeding + ruptured membranes + fetal distress = suspect vasa previa. (Fetal Emg)

PLACENTA ACCRETA SPECTRUM

Abnormal placental attachment where chorionic villi invade beyond decidua basalis
Placenta fails to separate after delivery

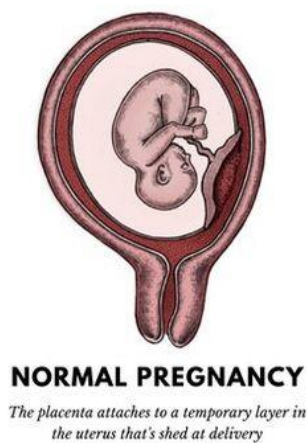
Types (Depth of Invasion)

Accreta → Attached to myometrium

Increta → Invades into myometrium

Percreta → Penetrates through uterus ± bladder

NORMAL PLACENTA VS. PLACENTA ACCRETA SPECTRUM (PAS)



PLACENTA ACCRETA

When the placenta attaches too deeply into the uterine wall



PLACENTA INCRETA

When the placenta attaches into the uterine muscle



PLACENTA PERCRETA

When the placenta goes completely through the uterine wall, sometimes invading nearby organs like the bladder

Risk Factors (Very High Yield)

- Previous cesarean section (most common)
- Placenta previa
- Prior uterine surgery or curettage
- Multiparity
- Advanced maternal age

Clinical Features

- Severe postpartum hemorrhage
- Retained placenta
- Failure of placenta to deliver
- Hypovolemic shock

Diagnosis

- Antenatal ultrasound with Doppler
- MRI
- Often suspected before delivery

Management (NORCET Priority)

- Do NOT attempt manual removal
- Planned cesarean hysterectomy
- Blood products prepared

Nursing Responsibilities

- Anticipate massive hemorrhage
- Monitor vital signs & blood loss
- Maintain IV access (large bore)
- Prepare for transfusion
- Emotional support & counseling

Complications

- Massive hemorrhage
- DIC
- Infection

- Injury to bladder/adjacent organs
- Maternal mortality

EXAM PEARLS

- Placenta does not detach → think ACCRETA
- Manual removal = fatal bleeding
- Strong association: placenta previa + previous C-section



PRENATAL SCREENING & DIAGNOSIS

1st Trimester Screening (11-14 Weeks)

✓ Ultrasound Markers

- ↑ Nuchal translucency (NT) (most important)
- Absent nasal bone
- LAb: normal ductus venosus flow
- Tricuspid regurgitation

✓ Serum Analytes (Dual Marker)

- β -hCG
- PAPP-A

In down syndrome:

Condition	β -hCG	PAPP-A
Trisomy 21	↑	↓
Trisomy 18	↓	↓
Trisomy 13	↓ / normal	↓

✓ **Nuchal Translucency (NT) - MUST KNOW**

- Measured at 11-13+6 weeks
- NT > 3 mm (>99th percentile) → significant
- ↑ NT associated with:
 1. Aneuploidy
 2. Cardiac defects
 3. Hydrops
 4. Cystic hygroma

2nd Trimester Screening (15-21 Weeks)

✓ **Triple Test**

- Free β -hCG
- AFP
- Unconjugated estriol (uE3)

✓ Quadruple Test

- Triple test + Inhibin A

Marker	Trisomy 21
β -hCG	↑
AFP	↓
uE3	↓
Inhibin A	↑

✓ NIPT / Cell-Free DNA

- Screening test (NOT diagnostic), FOR NURSES
- Can be done after 9-10 weeks
- Detection rate for Down syndrome: ~99%

PRENATAL DIAGNOSTIC TESTS

✓ Amniocentesis

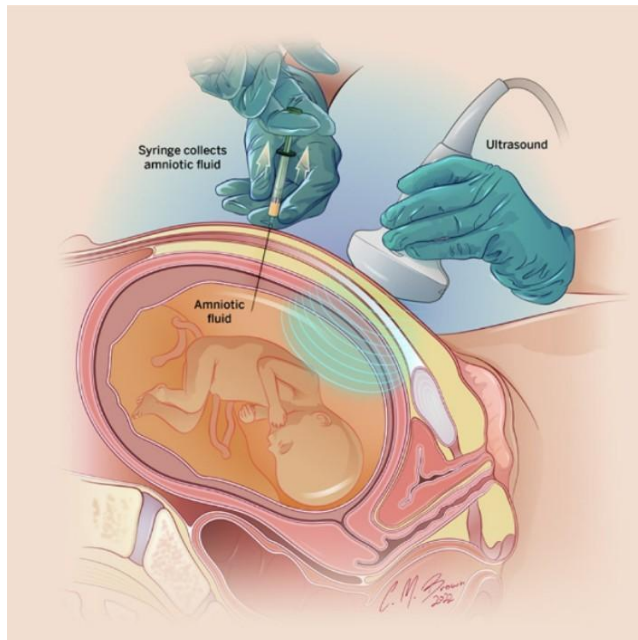
Done at 15–20 weeks

Indications:

- Genetic disorders
- Congenital infections

- Rh isoimmunization
- Fetal lung maturity

Complication risk: 0.1-0.3%

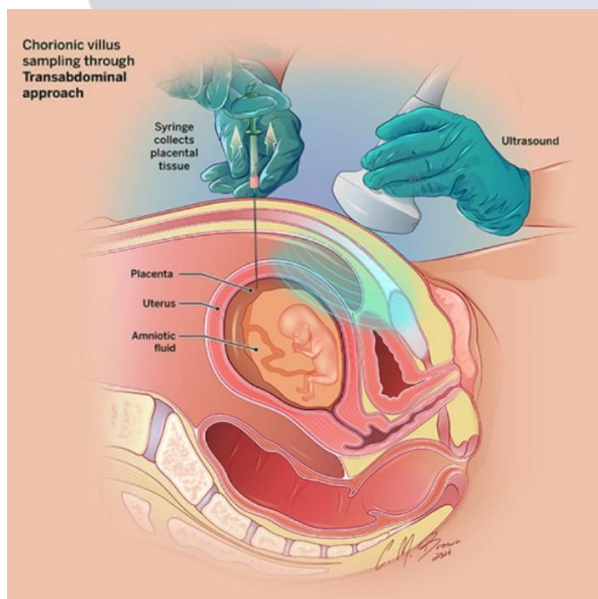


✓ Chorionic Villus Sampling (CVS)

Done at **10-13 weeks**

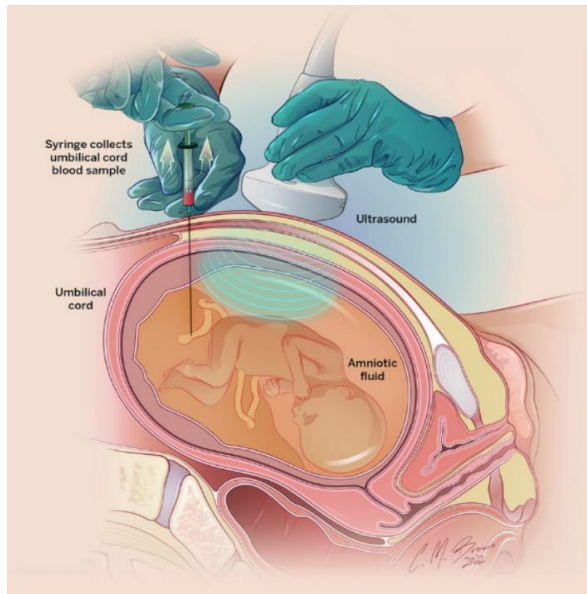
Advantage: Early diagnosis

Risk ↑ if done <7 weeks (limb defects)



✓ Cordocentesis

- After 18 weeks
- Used in Rh isoimmunization
- Allows intrauterine transfusion



NORCET PEARL

- Screening ≠ Diagnostic
- Abnormal screening → diagnostic test
- NIPT has highest screening accuracy



ABORTION AND EARLY PREGNANCY COMPLICATIONS

ABORTION:

Spontaneous Abortion (Miscarriage)

Definition: Termination of pregnancy before 20 weeks or fetal weight < 500g.

Clinical Classifications

Type	Bleeding	Cervical Os
Threatened	Mild	Closed
Inevitable	Moderate/Heavy	Open
Incomplete	Heavy	Open
Complete	Subsiding	Closed
Missed	Minimal/Brownish	Closed

Septic Abortion: Any abortion complicated by infection (fever > 100.4°F, foul discharge)

Priority: Broad-spectrum antibiotics and evacuation.

2. RECURRENT PREGNANCY LOSS (RPL)

Definition: 2 or more consecutive pregnancy losses (confirmed by USG/HPE).

Etiology & Treatment:

✓ Genetic: Balanced translocation (most common). Tx: Genetic counseling.

✓ Endocrine: Luteal Phase Defect (decreased progesterone), Uncontrolled DM, or Thyroid dysfunction.

✓ Anatomical: * *Cervical Insufficiency:* Painless 2nd-trimester loss. Tx: Cervical Cerclage (12-14 weeks).

Cervical Cerclage

Types: McDonald (most common), Shirodkar, transabdominal.

Contraindications (6 A's): Active bleeding, Active labor, Amnionitis, Anomalous fetus, Abruptio placentae, Advanced dilation (>4cm).

✓ **Uterine Anomalies:** Septate uterus (most common).

✓ **Immunological:** Antiphospholipid Antibody Syndrome (APLA).

Tx: Low-dose Aspirin + LMWH (started once cardiac activity is seen).

3. ECTOPIC PREGNANCY

Definition: Implantation outside the uterine cavity.

Most common site: Ampulla of the Fallopian tube.

Clinical Features (The "Classic Triad")

- Delayed menstruation (Amenorrhea).
- Abdominal pain (Unilateral or generalized).
- Vaginal bleeding (Usually spotting).

Key Diagnostics

- **hCG Discriminatory Zone:** If hCG is 1500-2000 IU/L and no gestational sac is seen on Transvaginal Ultrasound (TVS), suspect Ectopic Pregnancy.
- **USG Sign:** "Ring of Fire" (high vascularity around the adnexal mass).
- **Culdocentesis:** Aspiration of non-clotting blood from the Pouch of Douglas (confirms hemoperitoneum).

Management:

- ***Medical (Methotrexate):*** * Criteria: Hemodynamically stable, sac size < 3.5 cm, no fetal heart rate, hCG < 5000 IU/L.
Mechanism: Folic acid antagonist (blocks DNA synthesis in trophoblasts).
- ***Surgical: Salpingostomy***
Incision to remove the sac (preserves the tube).

- **Salpingectomy:** Removal of the entire tube (preferred if tube is ruptured or damaged).

NURSING "MUST-KNOWS"

- Anti-D (RhoGAM): Must be given to Rh-negative unsensitized mothers within 72 hours of any abortion or ectopic pregnancy to prevent isoimmunization.
- Cerclage Care: Monitor for signs of labor (contractions), infection, or ROM (leaking fluid). Cerclage is usually removed at 37 weeks.
- Ruptured Ectopic = Emergency: Monitor for hypovolemic shock (tachycardia, hypotension, shoulder pain due to diaphragmatic irritation).
- Culdocentesis: Aspiration of non-clotting blood from posterior cul-de-sac suggests hemoperitoneum. Largely replaced by USG.
- Heterotopic Pregnancy: Co-existing IUP and ectopic. Medical management is contraindicated; treatment is surgical.



MOLAR PREGNANCY

Abnormal trophoblastic proliferation after abnormal fertilization

Types

- Complete mole
- Partial mole

Genetics (MOST TESTED)

Complete Mole

- All paternal DNA
- Empty ovum
- 46XX (most common)
- No fetus

- p57 KIP2 - NEGATIVE

Partial Mole

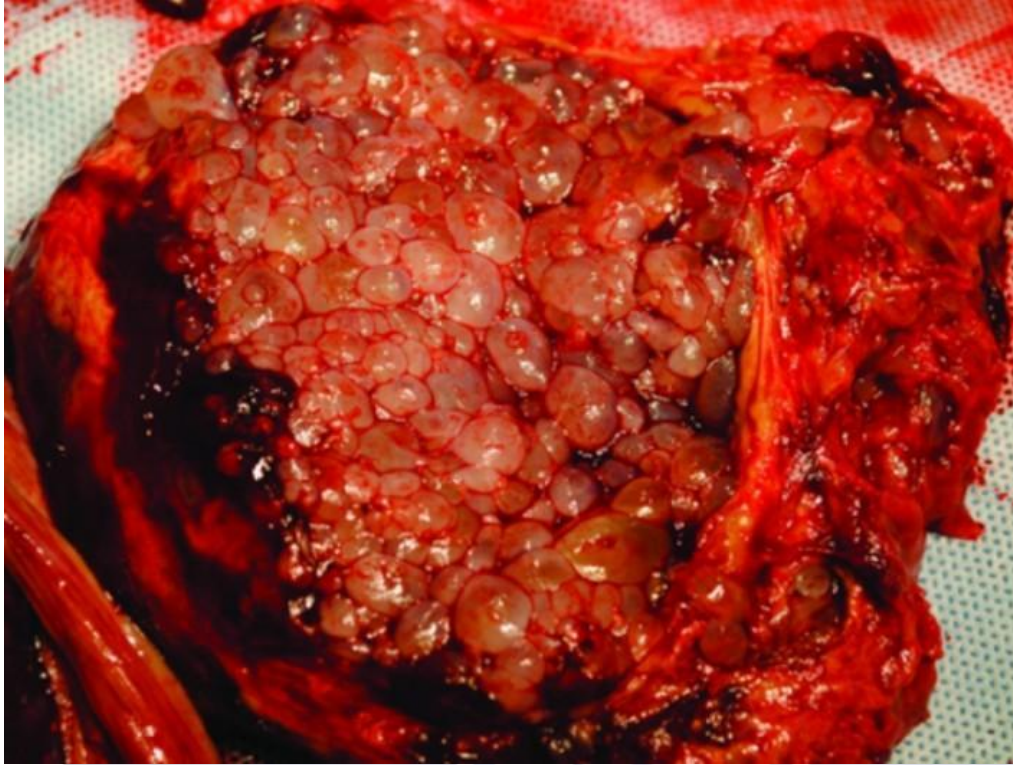
- Triploid
- 2 sperms + normal ovum
- 69XXX / 69XXY
- Fetal parts present
- p57 KIP2 - POSITIVE

Complete vs Partial (EXAM TABLE)

Feature	Complete	Partial
Fetus	Absent	Present
β -hCG	>100,000	<100,000
Villous edema	Diffuse	Focal
GTN risk	15–20%	1–5%
p57	Negative	Positive

Clinical Features

- Vaginal bleeding + grape-like vesicles



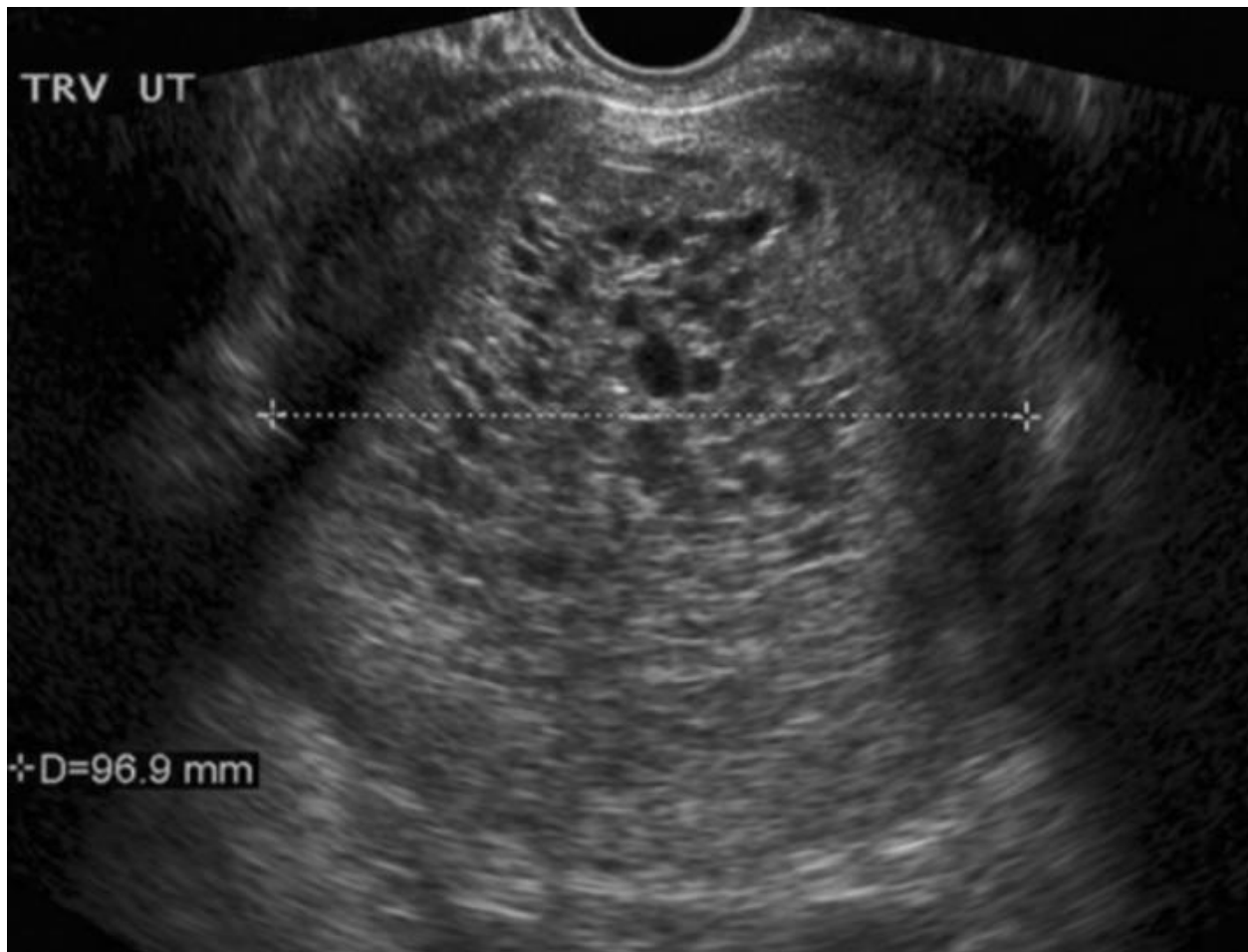
- Uterus > POG
- Hyperemesis
- Early pre-eclampsia (<20 weeks)
- Thyrotoxicosis

Diagnosis

- Very high β -hCG
- Urine test $\rightarrow$ hook effect
- USG $\rightarrow$ Snowstorm appearance

NURSES DUO

FOR NURSES



Management

- Suction & evacuation (Tx of choice)

Post-Evacuation Care

- Anti-D if Rh negative
- Serial β -hCG
 - 48 hrs → baseline
 - Weekly till negative
 - Monthly × 6 months
- Contraception mandatory

Risk Factors → GTN

- Complete mole
- ↑ maternal age

- β -hCG $>100,000$
- Uterus $>$ gestational age
- Theca lutein cyst >6 cm
- Slow fall of β -hCG

NORCET PEARLS

- Early preeclampsia = think mole
- p57 negative = complete mole
- Screening $\neq$ diagnostic
- Evacuation, not induction



INTRAPARTUM COMPLICATIONS

BY NURSES, FOR NURSES 

CORD PROLAPSE:

Cord Presentation: A loop of the umbilical cord is below the fetal presenting part while the membranes are still intact.

Cord Prolapse: A loop of the cord is below the presenting part after the membranes have ruptured. This is a critical OBSTETRIC EMERGENCY.

Pathophysiology & Risks

When the cord prolapses, it can become compressed between the fetus and the pelvic wall, or undergo vasospasm. This leads to:

- Acute fetal distress.
- Fetal hypoxia.
- Potential fetal death.

Diagnosis & Management

Diagnosis

The cord may be seen at the vaginal opening (introitus) or felt during a vaginal exam.

- Ultrasound can occasionally identify a cord presentation.

Immediate Emergency Management

The Goal: Relieve pressure off the cord immediately to restore fetal blood flow.

Positioning: Place the mother in Trendelenburg, Knee-chest (All 4s), or a Sims position.

Manual Displacement: A clinician should keep a hand in the vagina to manually push the fetal presenting part upward and away from the cord.

Bladder Filling: Instilling fluid into the bladder can help push the presenting part up.

GOLDEN RULE: NEVER touch or try to push the cord back in; this causes vasospasm.

Delivery Mode

Cesarean Section:

- Required for a live fetus
- The goal is delivery within 30 minutes (preferably 15 mins).

Vaginal Delivery:

- Only attempted if the cervix is fully dilated and instrumental delivery (forceps/vacuum) is immediate.

Dead Fetus:

- If fetal death is confirmed, a spontaneous vaginal delivery is allowed to avoid unnecessary maternal surgery.

Emergency Nursing Interventions: The "5 S's"

If you feel a pulsating cord during a vaginal exam or see it protruding:

Stay and Signal: Never leave the patient; call for help and notify the surgical team immediately for a Category 1 C-section.

Sterile Hand (Manual Displacement): Keep your gloved hand in the vagina and push the presenting part (e.g., the head) up and away from the cord to relieve pressure.

Shift Position: Move the mother into Knee-Chest or Trendelenburg position immediately to use gravity to shift the fetus off the cord.

Stop Touching: Do not attempt to push the cord back inside; handling the cord causes vasospasm, which further cuts off fetal oxygen.

Swift Delivery: Prepare for delivery within 30 minutes (ideally 15) if the fetus is alive; if the fetus has died, spontaneous delivery is the goal.

SHOULDER DYSTOCIA:

Definition: A condition where the fetal shoulders become stuck after delivery of the head; it is an emergency if delivery of the shoulder takes > 1 minute after the head.

Key Assessment & Diagnosis

Turtle Sign: The fetal head recoils back against the perineum after delivery.

Risk Factors:

- Maternal diabetes
- fetal macrosomia (> 4.5 kg)
- maternal obesity (BMI > 30 kg/m²)
- prolonged labor stages

Complications:

Fetal:

- Asphyxia
- Erb's palsy (C5,6)
- Klumpke palsy (C8-T1)
- humerus/clavicle fractures.

Maternal: Postpartum hemorrhage (PPH) and high-degree perineal tears.

Management: The HELPER Mnemonic

Nursing priority is to follow these steps in order:

- ✓ **H - Help:** Call for additional assistance and the surgical team.
 - ✓ **E - Evaluate for Episiotomy:** To provide more room for internal maneuvers
 - ✓ **L - Legs (McRoberts Maneuver):** First-line nursing action. Hyperflex the mother's legs onto the abdomen to flatten the lumbar spine and widen the pelvis.
 - ✓ **P - Pressure (Suprapubic):** Apply pressure just above the pubic symphysis (directed 45° downward) to push the anterior shoulder under the symphysis.
- Note: Never apply fundal pressure as it worsens the impaction.

✓ **E - Enter Rotational Maneuvers:**

Rubin II: Pressure on the posterior aspect of the anterior shoulder.

Woods' Screw: 180° rotation of the posterior shoulder.

✓ **R - Remove the Posterior Arm:** Manually deliver the posterior arm to reduce the shoulder diameter.

✓ **R - Roll the Patient:** Move the mother to her hands and knees (Gaskin Maneuver) to widen the pelvic outlet.

Critical Last-Resort Maneuvers

Zavanelli Maneuver: Pushing the fetal head back into the uterus followed by an emergency Cesarean section.

Intentional Clavicular Fracture: Breaking the fetal collarbone to reduce the shoulder width.

Prevention

Elective CS: Recommended by ACOG for non-diabetic women with a fetus > 5 kg or diabetic women with a fetus > 4.5 kg.

BREECH PRESENTATION

Definition: A longitudinal lie where the buttocks or lower limbs are the presenting part.

Types of Breech

- Frank Breech (65%): Flexed hips with extended knees.
- Complete Breech (10%): Flexed hips and flexed knees.
- Footling Breech: One or both feet are the presenting part; most common in preterm singleton breeches.
- Knee Presentation: Extended hip with a flexed knee.

Frank Breech (65%)	Complete Breech (10%)	Incomplete Breech (25%)	
		Footling Breech	Kneeling Breech
			
The baby's hip joints are flexed and knee joints are extended.	The baby's hip and knee joints are flexed.	The baby's hip and knee joints are extended on one or both sides.	The baby's hip joints are extended and knee joints are flexed on one or both sides.

Management Strategies

1. External Cephalic Version (ECV)

- Goal: Manually turning the baby to a cephalic presentation.
- Timing: Ideally performed at 36 weeks.
- Contraindications: Placenta previa, contracted pelvis, multiple pregnancy, and uterine anomalies.

2. Caesarean Section (C-Section)

- Indications: Primigravida, large fetus (> 3.75 kg), hyperextended head ("Star-gazing fetus"), or footling/incomplete breech.

3. Vaginal Breech Delivery

- Prerequisites: Frank breech, estimated weight < 3.75 kg, flexed head, and adequate pelvis.

Key Maneuvers:

- **Lovset Maneuver:** Used for delivery of the shoulders if they are arrested.
- **Mauriceau-Smellie-Veit Method:** Used for delivery of the after-coming head.
- **Burns-Marshall Method:** The fetus is left hanging so its weight exerts gentle downward traction for head delivery.

Critical Complications of Breech Delivery

Maternal Complications

- Prolonged labor and maternal distress.

- Lacerations (perineal, cervical, or vaginal).
- Postpartum hemorrhage (PPH).

Fetal Complications

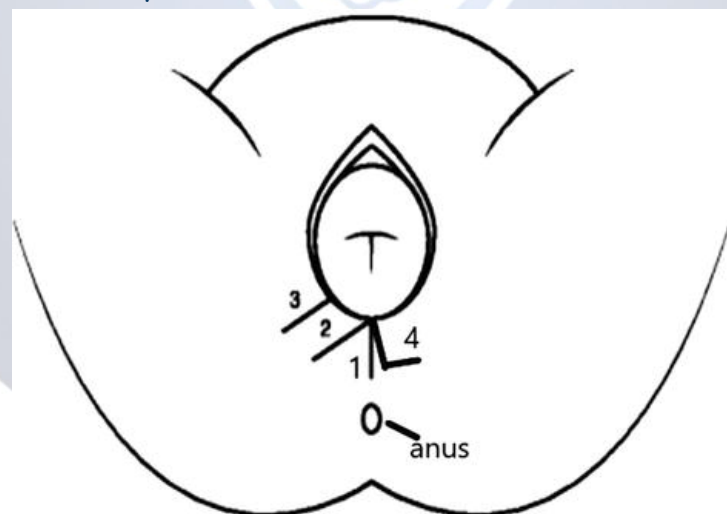
- Intracranial Hemorrhage: Common cause of death due to sudden compression/decompression of the head.
- Asphyxia: Risk from cord prolapse or compression.
- Fractures: Femur, humerus, or clavicle.
- Nerve Injuries: Brachial plexus injury or lacerations to the sternomastoid muscles.

EPISIOTOMY:

A deliberate incision made on the posterior vaginal wall and perineum during the second stage of labor to assist in the delivery of the fetus.

Key Clinical Concepts

- Timing: Performed at the time of crowning.
- Practice Standard: Routine episiotomy is NOT recommended; it should only be used restrictively.



Type of Episiotomy

1; Median, 2; Mediolateral, 3; Transverse
4; J-shaped

Indications:

- Rigid perineum (common in first-time mothers).

- Fetal macrosomia or shoulder dystocia.
- Instrumental deliveries (forceps or vacuum).
- Malpresentations (e.g., face to pubis, assisted breech).

Nursing Priorities (NORCET Tips)

- Risk Management: The nurse must monitor for hematoma formation post-procedure; ensuring no "dead space" is left during suturing is critical to prevent this.
- *Questions often prioritize post-op care like sitz baths and keeping the area clean and dry.*

Assessment: Always assess the repair site for redness, edema, ecchymosis, discharge, and approximation (REEDA scale).

PERINEAL TEAR

First-degree

- Involves: Skin only
- Perineal muscles not involved
- Usually heals easily, may not need suturing

Second-degree

- Involves: Skin + perineal muscles
- Anal sphincter NOT involved
- Most common tear, needs suturing

Third-degree

- Involves: Skin + perineal muscles + anal sphincter

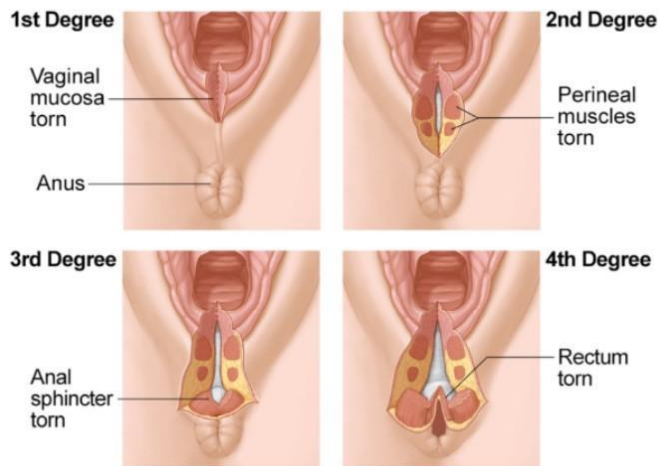
Subtypes:

- 3a: <50% external anal sphincter torn
- 3b: >50% external anal sphincter torn
- 3c: External + internal anal sphincter torn
- Needs surgical repair

Fourth-degree

- Involves: Skin + muscles + anal sphincter + rectal mucosa
- Most severe

- High risk of infection & incontinence



AMTSL:

Active Management of Third Stage of Labor (AMTSL) is a clinical protocol designed by the WHO in 2007 to prevent Atonic Postpartum Hemorrhage (PPH).

Core Components of AMTSL

According to the established guidelines, there are three primary steps:

- **Administering a Uterotonic:** Essential to stimulate uterine contractions.
- **Controlled Cord Traction (CCT):** Performed only if a skilled birth attendant is present.
- **Uterine Massage:** Sustained massage immediately following placental delivery is no longer universally recommended for all women; instead, frequent postpartum uterine tone assessments are preferred to identify uterine atony early.

Uterotonic Options

- **Oxytocin:** The gold standard choice (10 IU IM/IV).
- **Carbetocin:** 100 mcg IM/IV.
- **Misoprostol:** (400 or 600 mcg PO) used primarily where skilled attendants or refrigeration are unavailable.
- **Ergometrine/Methylergometrine:** (200 mcg IV) strictly avoided in patients with hypertension.

Fixed-dose Combinations: Oxytocin and ergometrine (5 IU/500 µg IM).

Benefits of AMTSL

- Decreased incidence of PPH.
- Shortened third stage of labor.
- Reduced need for blood transfusions and therapeutic uterotonic drugs.
- Decreased percentage of labors where the third stage exceeds 30 minutes.



LABOUR

Definition and Diagnosis of Labor

✓ True labor is characterized by:

Regular, painful uterine contractions that increase in frequency and intensity.
Cervical changes, specifically effacement (thinning) and dilation.

✓ Show: The expulsion of the cervical mucus plug mixed with a small amount of blood.

✓ The Four Stages of Labor

Stage 1: Cervical Stage

This stage begins with the onset of true labor pains and ends when the cervix is fully dilated (10 cm).

It is further divided into latent, active and transition stages

Latent Phase

- Cervical dilation: 0-3
- Contractions: Mild, irregular → regular
- Duration: Longest phase

Nursing focus:

- Comfort measures
- Encourage rest & hydration
- Teach breathing & relaxation

Active Phase

- Cervical dilation: 4-7 cm
- Contractions: Moderate-strong, regular
- Cervical change: Faster dilation

Nursing focus:

- Pain management
- Monitor fetal heart rate
- Emotional support

Transition Phase

- Cervical dilation: 8-10 cm
- Contractions: Very strong, frequent

Mother may feel:

- Urge to push
- Anxiety, nausea, irritability

Nursing focus:

- Continuous support
- Encourage controlled breathing
- Prepare for second stage

Key NORCET Points ☆

- First stage = longest stage
- Ends at 10 cm dilation
- Transition is most intense but shortest
- Do not encourage pushing until full dilation

Stage 2: Fetal Expulsion Stage

This stage begins at full cervical dilation and ends with the delivery of the fetus.

Propulsive Phase: From full dilation until the fetal head reaches the pelvic floor.

Expulsive Phase: Characterized by maternal "bearing down" efforts and ends with birth.

Duration: Typically lasts 1-2 hours in primigravida and 30 minutes in multigravida.

Nursing Responsibilities

- Encourage effective pushing with contractions
- Monitor fetal heart rate (before, during, after pushing)
- Support position changes
- Observe for perineal bulging & crowning
- Provide emotional coaching

Stage 3: Placental Stage

This stage begins after the birth of the baby and ends with the expulsion of the placenta and membranes.

Duration: Usually lasts 5-15 minutes.

Signs of Separation: Gush of blood, lengthening of the cord, and the uterus becoming firm and globular.

Management: Can be expectant or involve Active Management of Third Stage of Labor (AMTSL) to prevent hemorrhage.

Nursing Care

- Administer oxytocin after delivery (as prescribed)
- Observe for excessive bleeding

- Inspect placenta for completeness
- Do not pull the cord (prevents uterine inversion)

NORCET Pearls ☆

- Greatest risk of postpartum hemorrhage begins here
- Placenta should be delivered intact

Stage 4: Observation Stage

This is the first hour following placental delivery.

Clinical Importance: This is a critical time to monitor the mother for Postpartum Hemorrhage (PPH).

Focus: Maternal stabilization & bonding

Assessment Priorities

- Fundus: Firm, midline
- Lochia: Rubra, moderate
- Vital signs: Watch for hypotension/tachycardia
- Bladder: Full bladder → uterine atony

Nursing Care

- Massage uterus if boggy
- Encourage breastfeeding (↑ oxytocin → ↓ bleeding)
- Promote skin-to-skin contact
- Manage pain & provide warmth

NORCET Pearls ☆

- Boggy uterus = uterine atony
- First action → fundal massage
- Excessive bleeding → suspect postpartum hemorrhage

Mechanism of Labor (The Seven Cardinals)

For the fetus to pass through the pelvis, it undergoes these movements:

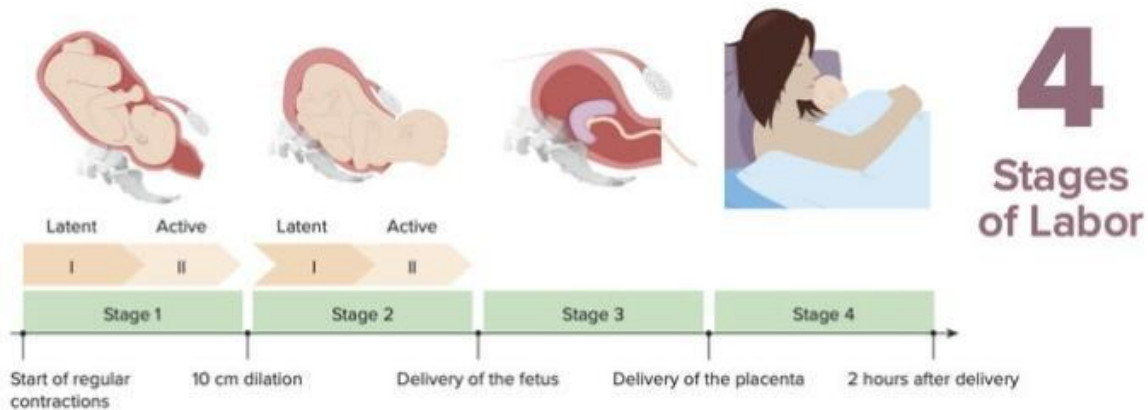
- ✓ Engagement: Fetal presenting part enters the pelvic inlet.
- ✓ Descent: Continuous downward movement through the birth canal.
- ✓ Flexion: Fetal chin touches the chest to present the smallest head diameter.
- ✓ Internal Rotation: Head rotates to fit the pelvic outlet.
- ✓ Extension: The head is delivered as it pivots under the symphysis pubis.
- ✓ Restitution/External Rotation: The head aligns with the shoulders.
- ✓ Expulsion: Delivery of the shoulders and the rest of the body.



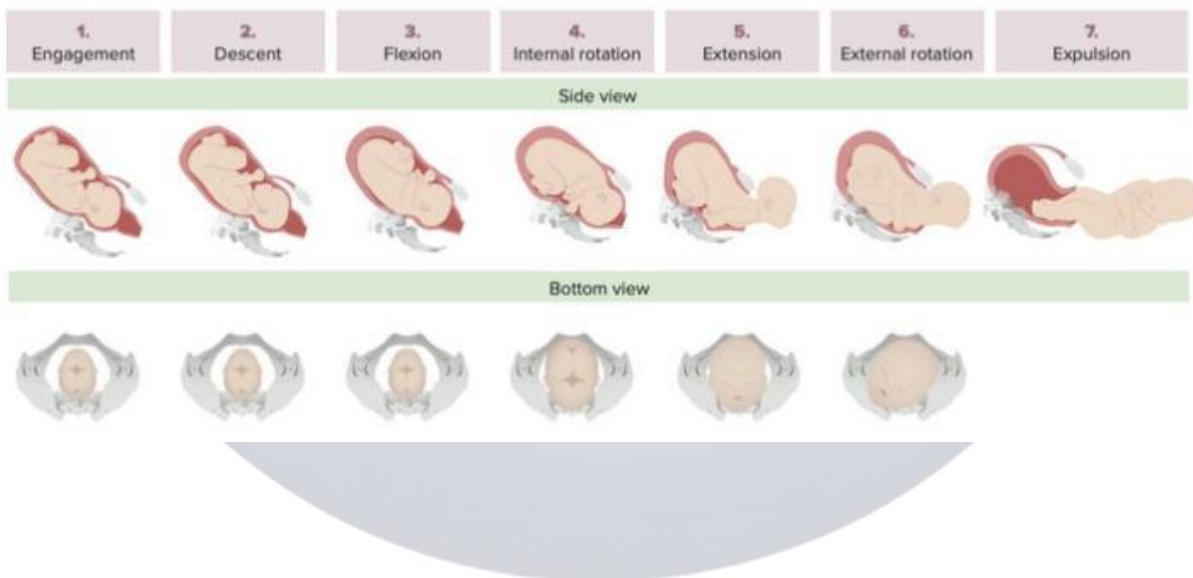
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MECHANISMS & STAGES OF LABOR



7 Cardinal Mechanisms of Labor





PUERPERIUM

Definition: It is the period following childbirth during which the body's tissues, especially the pelvic organs, revert back to their pre-pregnant state both anatomically and physiologically.

Duration: This period begins as soon as the placenta is expelled and traditionally lasts for **6 weeks (42 days)**.

Clinical Subdivisions:

- Immediate: Within the first 24 hours.
- Early: From 2 days to 7 days.
- Remote: From 1 week to 6 weeks.

Physiological Changes: The Uterus & Lochia

The most critical assessment during the puerperium is involution, the process of the uterus returning to its normal size.

Uterine Involution

- ✓ **Immediately after delivery:** The uterus is at the lower border of the umbilicus (approx. 20 weeks size).
- ✓ **Day 1:** 1 finger breadth below the umbilicus.

- ✓ Day 2: 2 finger breadths below the umbilicus.
- ✓ End of 2 weeks: No longer palpable abdominally; it becomes a pelvic organ.
- ✓ End of 6-8 weeks: Returns to its pre-pregnant size.

Lochia (Vaginal Discharge)

Lochia consists of blood, deciduas, and mucus. It follows a specific timeline:

- ✓ **Lochia Rubra (Red)**: Days 1-4.
- ✓ **Lochia Serosa (Yellow-brown)**: Days 5-9.
- ✓ **Lochia Alba (White)**: Days 10-28.

NORCET Tip: If a patient reports Lochia Rubra on Day 10, this is abnormal and may indicate sub-involution due to retained placental fragments or infection.

2. POSTPARTUM COMPLICATIONS

Puerperal Pyrexia & Sepsis

Pyrexia: A temperature of 38°C (100.4°F) or higher occurring after the first 24 hours on 2 occasions within 24 hours.

- Sepsis: Often caused by **E. coli**.
- Symptoms include pelvic pain, fever, and foul-smelling discharge.
- Management: Broad-spectrum antibiotics (Ampicillin + Metronidazole + Gentamicin).

Breast Complications

Mastitis:

- Most common organism is *Staphylococcus aureus*.
- Presents as a high fever with a swollen, red, tender breast
- Treatment involves antibiotics (Dicloxacillin) and continuing to breastfeed to empty the breast.

Breast Abscess: Requires surgical drainage (deep radial incision).

3. Psychiatric Disorders in the Puerperium

	Postpartum Blues	Postpartum Depression	Postpartum Psychosis
Incidence	35-70%	10-15%	0.1%
Time of onset	2-4 days PP	2 weeks-12 months PP	2-3 days PP
Duration	Resolves within 10 days	13-14 months	Variable
Symptoms	Insomnia, tearfulness, fatigue, irritability, poor concentration	Irritability, labile mood, difficulty falling asleep, sad mood, phobias, anxiety	Mania and/ or mixed affective state, agitation, delusions, disorganized behavior, infanticide
Risk Factors	Very common	Adverse delivery experience, Still birth, H/o depression, adverse childhood experience, recent life events, lack of social support	Past/ho psychosis, H/o bipolar or family H/o bipolar ds
Treatment	Self limiting	CBT + Antidepressants	In patient, Pharmacotherapy

INFECTION IN PREGNANCY

1. HIV IN PREGNANCY

(Key points kept concise for revision)

- Vertical transmission without ART: ~30%
- With ART: ↓ to 1-2% ✓

Most transmission: Intrapartum

Diagnosis

- ELISA → confirm Western blot

Management

- TDF + 3TC + EFV (lifelong)
- C-section only for obstetric indication
- Avoid invasive procedures

Postpartum

- Exclusive breastfeeding × 6 months
- No mixed feeding

Neonate

- Nevirapine / Zidovudine × 6 weeks

2. RUBELLA IN PREGNANCY

Most dangerous in 1st trimester

Diagnosis

- IgM positive → recent infection

Congenital Rubella Syndrome (CRS) Triad (VERY HIGH YIELD):

- PDA
- Cataract
- Sensorineural hearing loss

Prevention

- MMR vaccine before or after pregnancy
- Avoid pregnancy for 1 month post-vaccination

3. VARICELLA (CHICKENPOX)

Maternal risk

- Varicella pneumonia

Fetal effects

- Limb hypoplasia
- Skin scarring
- Eye defects

Management

VZIG for exposed, non-immune pregnant woman

✗ No role once rash appears

4. HERPES SIMPLEX VIRUS (HSV)

Primary risk: Intrapartum transmission

- Active genital lesions → C-section
- Neonatal herpes = high mortality

5. PARVOVIRUS B19 (ERYTHEMA INFECTIOSUM) (PYQ)

NORCET PEARL

☞ Causes fetal anemia, NOT malformations

Transmission

- Respiratory droplets

Fetal Effects

1. Severe fetal anemia
2. Hydrops fetalis
3. IUFD

Diagnosis

- Maternal IgM antibodies
- Fetal anemia → Doppler (↑ MCA velocity)

Management

- Close ultrasound monitoring
- Intrauterine transfusion if severe anemia

6. SYPHILIS IN PREGNANCY

Transmission

- Transplacental (any trimester)

Fetal Effects (VERY HIGH YIELD)

- Early congenital syphilis
- Snuffles
- Rash
- Hepatosplenomegaly

Late congenital syphilis

- Hutchinson teeth (PYQ NORCET)
- Saddle nose
- Deafness

Diagnosis

- Screen ALL pregnant women
- VDRL / RPR → confirm TPHA / FTA-ABS

Treatment

- Penicillin is the ONLY effective treatment ✓
- Treat mother and partner

⚠ Jarisch-Herxheimer reaction:

Fever, uterine contractions → fetal distress

7. COVID-19 IN PREGNANCY

Maternal Risks

- Severe disease
- Thromboembolism
- Respiratory failure

Fetal Risks

- Preterm birth
- Fetal distress
- Stillbirth (severe maternal illness)

Transmission

- Vertical transmission: rare

Management

- Treat similar to non-pregnant adults
- Oxygen, fluids, anticoagulation if indicated
- Vaccination recommended in pregnancy ✓
- Breastfeeding
- Safe with mask + hand hygiene

OPERATIVE OBSTETRICS

INSTRUMENTS IN OBSTETRICS & GYNAECOLOGY:

Sim's speculum:



Cusco's speculum:



Anterior Vaginal Wall retractor:



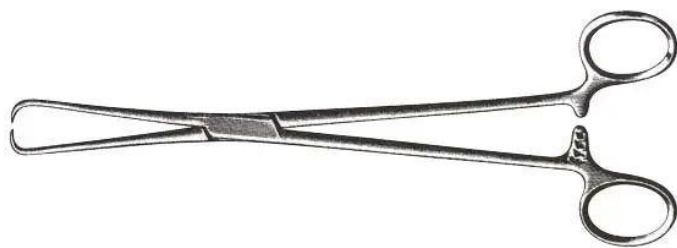
Vulsullem:



Uterine Sound:



Tenaculum:



Hegar dilator:

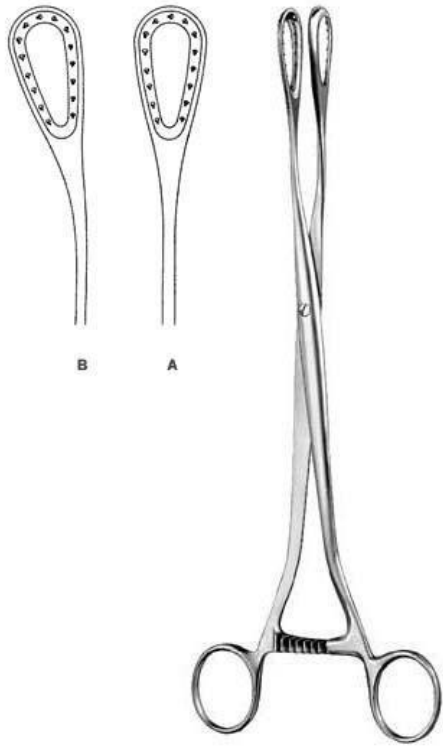


Manual Vacuum Aspirator and Karman cannula:



Ovum forceps:

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Green Armytage forceps:



Doyen's Retractor:



Episiotomy scissors:



Pap smear kit:

Pap Smear Kit



The PAP Smear Test Kit is a device that will simplify the process of obtaining a cervical PAP smear sample for cytological examination. It simplifies and maximizes cell sample collection for a more accurate examination. The Kit comprises a glass slide, wooden spatula, cotton-tipped applicator, and endocervical brush.



Sponge holding forceps:



Babcock's forceps:



Allis forceps:



Bonney myomectomy clamp:



Veress needle:



Trocar cannula:



Hysteroscope:



DUO
SES

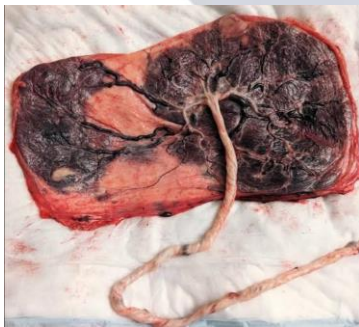


PLACENTAL ABNORMALITIES

Battledore Placenta



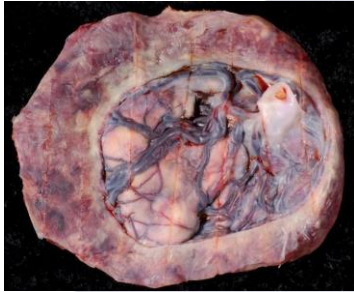
Placenta succenturiata



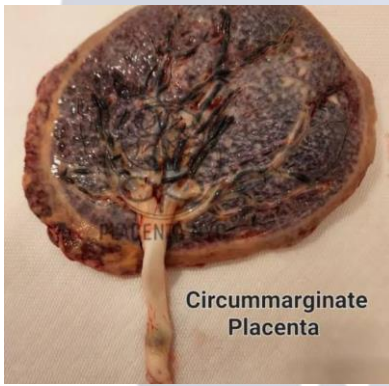
Placenta Bilobata



Circumvallate Placenta

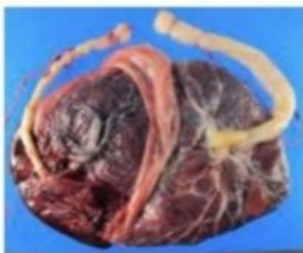


Circummarginate Placenta



Circummarginate
Placenta

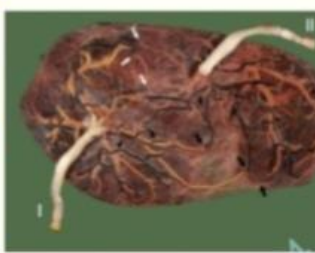
Twin Pregnancy



DCDA (Thick membrane)



MCDA (thin membrane)



MCMA (no membrane)



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